Pengzhen Li

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Summary

Economics Ph.D. candidate specializing in causal inference, structural estimation, and large-scale optimization, with four peer-reviewed publications, one with **1,000+ citations**. U.S. permanent resident; no sponsorship required.

Education

Ph.D. in Economics, University of Tennessee, Knoxville, TN — Expected May 2027
M.A. in Finance, Shandong University, Jinan, China — 2018
B.A. in Economics, Tongji University, Shanghai, China — 2015

Experience

Doctoral Researcher, Department of Economics, University of Tennessee — 2023 – Present
Independent research on trade policy using the universe of Chinese customs records (2000–2014) and tariff data for 123 countries, in Stata, Python and GAMS; three working papers below.

Graduate Research Assistant, University of Tennessee (USDA- and FAA-funded projects) — 2021 – 2023

[1] Built stochastic mixed-integer optimization models (GAMS) of aviation-fuel supply chains across 1,374 U.S. counties.

[2] Ran facility-siting, regional-impact (IMPLAN) and risk (@RISK) analyses for federal sponsors.

[3] Built reproducible ArcGIS/R/Python pipelines for high-resolution spatial data.

Working Papers

[1] **Selling Down-Market: Market Access and the Quality of Chinese Exports** (job market paper). Comparing the same firms' U.S. and non-U.S. sales around the 2001 U.S. grant of permanent normal trade relations to China, export quality fell about 4.5% more in highly exposed products.

[2] **Trade Restrictiveness Indices with Discriminatory Tariffs** (with J. Lake). Extends the index to preferential tariffs; in 2018 preferences cut the median average tariff by 16% but the index by only 5%.

[3] **The Impact of Mayoral Endorsements on Presidential Election Outcomes: Insights from Brazil** (with M. Van Essen and L. Lima). A constrained maximum-likelihood estimator for an endorsement game recovers the structural parameters to within 1% in Monte Carlo experiments. **2024 Holly Award for best second-year paper.**

Publications

[1] **Li, P.**, Yu, T. E., et al. (2026). Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability. *Biomass and Bioenergy*, 208, 108793.

[2] **Li, P.**, Yu, T. E., et al. (2024). Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA. *Transportation Research Record*, 2678(9), 639–654.

[3] Martinez, C., **Li, P.**, et al. (2023). Beef price spread relationship with processing capacity utilization. *Journal of the Agricultural and Applied Economics Association*, 2(1), 133–145.

[4] Du, K., **Li, P.**, and Yan, Z. (2019). Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data. *Technological Forecasting and Social Change*, 146, 297–303. **1,000+ citations**

Skills

Programming: Python (pandas, NumPy, scikit-learn), R, Stata, SQL, GAMS, MATLAB, ArcGIS, Git
Methods: causal inference (DiD, event study, IV, RDD, synthetic control); structural estimation; panel and time-series econometrics; forecasting; mixed-integer and stochastic optimization; machine learning
Other: Passed CFA Level II; English (fluent), Mandarin (native)

Awards

Charles B. Garrison Award for Excellence in Teaching, University of Tennessee — 2026
Summer Graduate Assistantship, University of Tennessee — 2025
J. Fred and Wilma Holly Award for best second-year paper, University of Tennessee — 2024
J. Fred and Wilma Holly Fellowship, University of Tennessee — 2023 – Present
Academic Scholarship for Master's Students and Freshman Scholarship, Shandong University — 2016 – 2017
National Endeavor Scholarship; Tongji Scholarship of Excellence, Tongji University — 2014